# Latency Arms Race



*Microseconds as competitive edge*

## PROFESSIONAL DEFINITION

The Latency Arms Race describes the competitive pursuit of ever-lower trading latency, where microseconds of speed advantage translate into profit, driving heavy investment in infrastructure.

## WHO DEVELOPED IT

A phenomenon documented in market-microstructure and high-frequency-trading research.

## WHY IT WAS DEVELOPED

It is described to explain the economics and risks of speed competition in modern markets.

## FIGURE 1 · LATENCY ARMS RACE

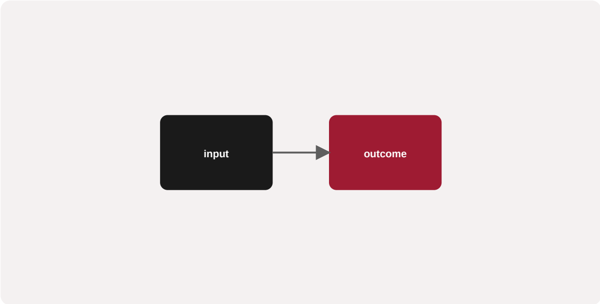


*Figure 1. A schematic of the Latency Arms Race as applied in BARBM312 (illustrative).*

## HOW IT IS USED

The value of marginal speed is weighed against its cost and systemic risk; participants decide how far to compete on latency.

## WHEN IT IS USED

When analysing high-frequency trading strategy and market structure.

## BY WHOM IT IS USED

Trading-firm strategists, regulators and researchers.

## INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It captures how speed itself became a contested, costly source of edge, with systemic implications.

## RELATED FRAMEWORKS IN THIS COURSE

- **Algo-Trading Pipeline** · Slibrary 24
- **Market Microstructure Map** · Slibrary 24
- **Flash-Crash Risk Loop** · Slibrary 24
- **Execution Cost Frontier** · Slibrary 24

## SOURCES (HARVARD REFERENCING STYLE)

Budish, E., Cramton, P. and Shim, J. (2015) 'The high-frequency trading arms race', Quarterly Journal of Economics, 130(4).

MacKenzie, D. (2021) Trading at the Speed of Light. Princeton: PUP.